

Correlation Analysis

First, I created a correlation matrix between changes in log GDP per capita, log health expenditure per capita, % of the population age 65+, and the death rate to see what predictors have a strong linear relationship with the dependent variable (death_change).

	death_change	log_gdp_change	log_health_change	\
death_change	1.000000	-0.077542	-0.006860	
log_gdp_change	-0.077542	1.000000	0.781329	
log_health_change	-0.006860	0.781329	1.000000	
pop_65_change	0.614353	0.008515	0.107532	

	pop_65_change
death_change	0.614353
log_gdp_change	0.008515
log_health_change	0.107532
pop_65_change	1.000000

There is an approximate correlation of -0.08 between death_change and log_gdp_change. This means there is a negative relationship between these 2 variables. As log_gdp_change is positive (GDP growth), then death_change is negative (decline in mortality rates). This aligns with the findings in graph 3. But this correlation is low, so the linear relationship between these variables is rather weak. This weak correlation is further supported by graph 5A, which displays a flat, downward-sloping line, and the majority of data points are randomly scattered.

Next, there is an approximate correlation of -0.007 between death_change and log_health_change. This means there is a negative relationship between these 2 variables. As log_health_change is positive (health expenditure growth), then death_change is negative (decline in mortality rates). This makes sense because logically, spending more on health should reduce the chances of death. But the correlation is negligibly low (~0), so the linear relationship between these variables is extremely weak. This extremely weak correlation is also supported by graph 5B, which displays a flat trend line with randomly scattered data points. Consequently, a change in the log of health expenditure per capita is likely a bad predictor, and we will not incorporate it in the model or run regressions on it.

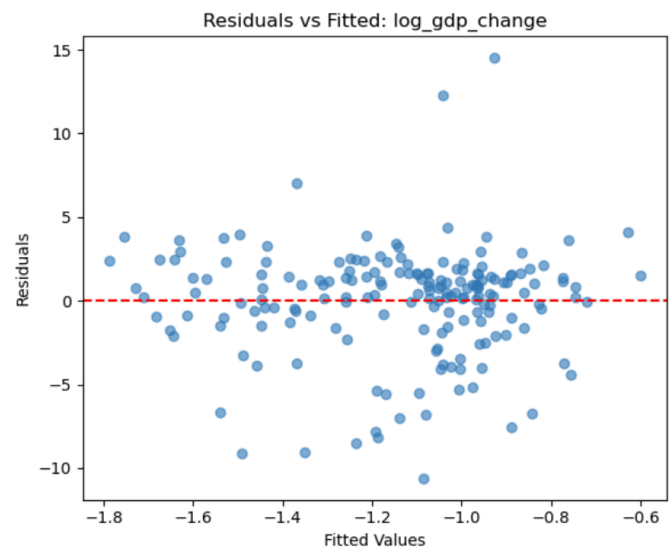
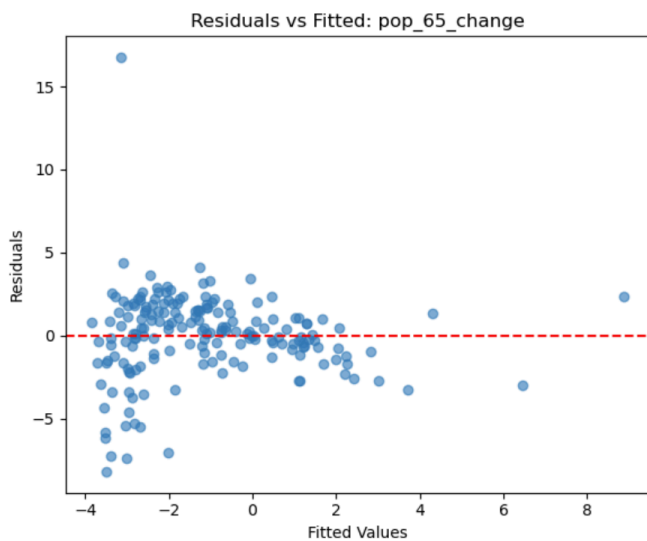
Lastly, there is an approximate correlation of 0.61 between death_change and pop_65_change. This means there is a positive relationship between these 2 variables. As pop_65_change is positive (growth in the elderly population), then death_change is also positive (growth in mortality rates). This makes sense because elderly people have weaker immunity, so elderly people are more likely to die, thereby leading to a higher mortality rate. Pop_65_change has the highest correlation with death_rate compared to the other variables. This strong correlation is

further supported by graph 5C, which displays an upward-sloping trend line with the majority of data points clustered near it.

Assumptions & Methodology

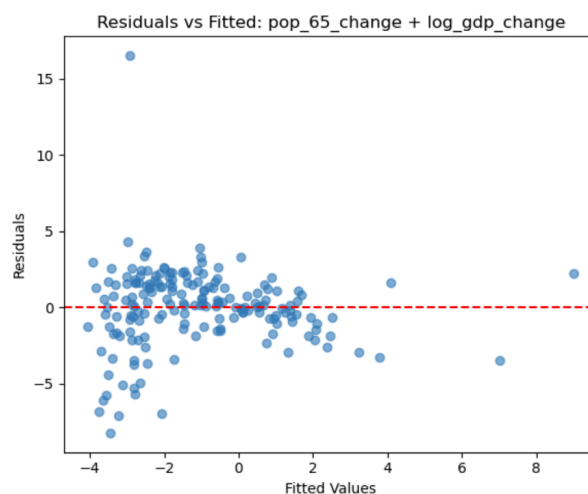
First, we establish that OLS regression uses some underlying assumptions: normally distributed residuals, constant variances for residuals (no heteroskedasticity), and a linear relationship truly exists between the data. I graphed residual plots for each predictor variable against the change in death rate to ensure that these conditions are satisfied.

The first residual plot shows the relationship between the change in % of the population age 65+ and the change in the death rate. The second residual plot shows the relationship between log_gdp_change and death_change. The third residual plot shows the relationship between log_gdp_change and pop_65_change against death_change (multivariate regression with 2 predictors). The first and third residual plots show non-random, tilted U-shaped patterns, and the second residual plot shows a funnel-like pattern. So, there is likely a non-linear relationship between the predictors and the change in mortality rates.



Residuals vs Fitted: log_gdp_change

Residuals vs Fitted: pop_65_change + log_gdp_change



For linear regressions with a single predictor, I determined the predictor's importance based on the p-values and R-Squared. Adjusted R-Squared, which penalizes for model complexity and additional terms, becomes more important for models with multiple predictors. For polynomial regressions, I determined the importance of the predictors based on the p-value of the higher-order term and the improvement in adjusted R-Squared. I also omitted writing out coefficients for polynomial models because the linear and squared terms are correlated with each other, and this makes individual coefficient interpretation misleading.

I wrote out the adjusted R-Squared for simple linear models because this allows me to compare adjusted R-Squared values across the simple linear, polynomial, and multivariate regressions to consistently measure these models' predictive power.

For p-values, we use a 5% level of statistical significance; p-values less than 0.05 are statistically significant, and any values higher than this are not statistically significant. A greater than 0.05 improvement in adjusted R-Squared is meaningful, and any improvement less than this is not meaningful. I specifically look for improvements in adjusted R-Squared, not R-Squared, because R-Squared always improves if we add more predictors, and it fails to account for model complexity and extra noise generated. So, assessing model fit based on adjusted R-Squared is more rigorous and accurate.

Regression Analyses (6 models)

Below is a table summarizing the important values for each of the regression models. After this table, we will analyze and compare the regressions in detail.

Regression #	Predictors	Coefficient	P-value	R ²	Adj. R ²
1	log_gdp_change (simple linear)	-0.4961	0.298	0.006	0
2	log_gdp_change_squared (polynomial)	_____	• 0.081 (log_gdp_change) • 0.122 (log_gdp_change_squared)	0.019	0.008
3	pop_65_change (simple linear)	0.8496	0	0.377	0.374
4	pop_65_change_squared	_____	• 0 (pop_65_change) • 0.159 (pop_65_change_squared)	0.384	0.377
5	log_pop_65_change (logarithmic)	0.6407	0	0.302	0.298
6	pop_65_change + log_gdp_change (multivariate linear)	• 0.8505 (pop_65_change) • -0.5296 (log_gdp_change)	• 0 (pop_65_change) • 0.16 (log_gdp_change)	0.384	0.377

Regression 1 Analysis (simple linear with log_gdp_change):

The coefficient of -0.4961 for log_gdp_change is economically intuitive because as the change in log(GDP per capita) increases by 1 unit (cumulative GDP per capita growth by 172% over 2000-2019), then the change in mortality rates decreases by 0.4961 percentage points. This aligns with my earlier discussion about how economic development promotes better health outcomes.

Note: A 1 unit increase in log(GDP per capita) corresponds to cumulative GDP per capita growth by 172% over 2000-2019, because if we did $\log(\text{GDP per capita 2019}) - \log(\text{GDP per capita 2000}) = 1$, that is equivalent to $\log(\text{GDP per capita 2019} / \text{GDP per capita 2000}) = 1$ via

logarithm rules. Then, GDP per capita 2019 / GDP per capita 2000 = $e = 2.718$, which is roughly 172% cumulative GDP per capita growth.

But the p-value of 0.298 is very high. So, log_gdp_change is NOT a reliable predictor because it is not statistically significant at the 5% threshold. The R-Squared is 0.006, so only 0.6% of the variation in changes in death rates is explained by changes in log(GDP per capita).

The adjusted R-Squared is 0, which means that none of the variation in death changes is explained by changes in log(GDP per capita), after penalizing for model complexity/additional terms.

Therefore, given the high p-value and low R-Squared, this means that log_gdp_change is likely an unreliable and weak predictor of changes in mortality rates.

Regression 2 Analysis (polynomial with log_gdp_change_squared):

Since the 2nd residual plot had a non-random pattern, I tried running a polynomial regression with log_gdp_change_squared vs. death_change (2nd order polynomial).

The R-Squared is 0.019, which means only 1.9% of the variation in changes in death rates is explained by the linear and squared terms for change in log GDP per capita. This is NOT meaningfully higher than the R-Squared value of 0.006 for the linear regression with log_gdp_change (only increased by 0.013). The adjusted R-Squared is 0.008, which means only 0.8% of the variation in changes in death rates is explained by changes in the linear and squared terms for change in log GDP per capita, after penalizing for model complexity/additional terms. This is NOT meaningfully higher than the adjusted R-Squared value of 0 for the simple linear regression with log_gdp_change.

The p-value for log_gdp_change is 0.081, and this is a big decrease relative to the p-value of 0.248 for the linear regression. But the first-order term is still not statistically significant. The p-value for log_gdp_change_squared is 0.122, so the squared term is not statistically significant either.

Since the squared term is not statistically significant and adding it did not significantly increase the adjusted R-Squared, it is unnecessary to use a polynomial regression. Given that log_gdp_change was an unreliable predictor for both simple linear and polynomial regressions, it will not help us predict changes in mortality rates.

Regression 3 Analysis (Simple linear regression for pop_65_change vs. death_change):

The coefficient for pop_65_change is 0.8496. This is economically intuitive because as the change in the proportion of the elderly population increases by 1 percentage point, then the change in mortality rates increases by 0.8496 percentage points, and it aligns with my earlier findings from graph 4.

The p-value is 0, which is very low. So, pop_65_change is a reliable predictor because it is statistically significant at the 5% threshold.

The R-Squared is 0.377, which means that 37.7% of the variation in changes in death rates is explained by changes in the % of the population that is elderly. The adjusted R-Squared is 0.374, which means that 37.4% of the variation in changes in death rates is explained by changes in the % of the population that is elderly, after penalizing for model complexity/additional terms.

Therefore, given the low p-value and relatively high R-Squared (compared to log_gdp_change), this means that pop_65_change is likely a reliable and strong predictor of changes in mortality rates.

Regression 4 Analysis (Polynomial, pop_65_change_squared vs. death_change):

Although the regression results suggest pop_65_change is reliable, we might obtain stronger results if we run a polynomial regression since the 1st residual plot showed a non-random pattern.

The R-Squared is 0.384, which means that 38.4% of the variation in changes in death rates is explained by the first and second order terms for changes in the % of the elderly population. This is NOT meaningfully higher than the R-Squared of 0.377 for the simple linear regression with pop_65_change (only 0.007 increase). The adjusted R-Squared is 0.377, which means that 37.7% of the variation in changes in death rates is explained by the first and second order terms for changes in the % of the population that is elderly, after penalizing for model complexity/additional terms. This is NOT meaningfully higher than the adjusted R-Squared of 0.374 for the simple linear regression with pop_65_change (only 0.003 increase).

The p-value for pop_65_change is 0, so the first-order term (pop_65_change) remains statistically significant. But, the p-value for pop_65_change_squared is 0.159, so the second-order term is NOT statistically significant.

Since the squared term is NOT statistically significant and including it did NOT meaningfully improve adjusted R-Squared, the polynomial model did not explain meaningfully more variation in mortality rate changes compared to the simple linear model. Thus, it is best not to use a polynomial regression for pop_65_change.

Regression 5 Analysis (Logarithmic, log_pop_65_change vs. death_change):

Although a polynomial regression did not add much value, I noticed that the scatterplot hinted at a logarithmic relationship - data points clustered on the left, steep upward movement, and curve flattening as we move rightward.

The coefficient for `log_pop_65_change` is 0.6407. This makes sense because as the log of the change in the proportion of the elderly population increases by 1%, then the change in mortality rates increases by 0.006 percentage points, and it aligns with my earlier findings from graph 4 and the simple linear regression.

Note: A 1% increase in the proportion of the elderly population means an increase of 0.01 for the $\log(\text{change in proportion of the elderly population})$ since $\log(1.01) = 0.01$. Then, the change in death rate is $0.6407 \times 0.01 = 0.006407$ percentage points.

The p-value is 0, so the predictor is statistically significant.

The R-Squared is 0.302, which means that 30.2% of the variation in changes in death rates is explained by changes in the log of % of the population that is elderly. This is meaningfully smaller than the R-Squared of 0.377 for the simple linear regression (0.075 decrease)! The adjusted R-Squared is 0.298, which means that 29.8% of the variation in changes in death rates is explained by changes in the log of % of the population that is elderly, after penalizing for added terms/model complexity. This is meaningfully smaller than the adjusted R-Squared of 0.374 for the simple linear regression (0.076 decrease)!

Although the predictor is statistically significant, the logarithmic regression actually explains much less variation in death rate changes than the simpler linear model does. Thus, we will not use this predictor variable to analyze changes in mortality rates.

Another important point is that the input values cannot be less than or equal to 0 for logarithms. Based on the code, countries with no or negative population changes (a decrease in the proportion of elderly populations) are forced to have a population change of 0.001% (infinitesimally small growth that is near 0% growth). This means that a country with a 3% decrease in the proportion of the elderly population is treated the same as a country with no change in the proportion of the elderly population, and this does not accurately represent the data.

So, logarithms fail to realize that in reality, population changes can be positive, negative, or 0. This is unlike variables such as GDP per capita and health expenditure per capita, where log transformations are effective because changes are strictly positive. Thus, another reason we will not accept logarithmic regression on the predictor is that doing this distorts the real population change data (which can be negative), thereby failing to illustrate the true relationship between population change and mortality rate changes.

Regression 6 Analysis (Multivariate, `pop_65_change` & `log_gdp_change` vs. `death_change`):

Lastly, I ran a multivariate regression between the 2 variables that had the highest correlation with changes in mortality rates. Given the previous analyses on the unreliability and

insignificance of `log_gdp_change`, I anticipate that adding this predictor will not substantially improve the model's predictive power.

The coefficient for `pop_65_change` is 0.8505. This means that holding the change in $\log(\text{GDP per capita})$ constant, when the change in the proportion of the elderly population in a country increases by 1 percentage point, then the change in the mortality rate increases by 0.8505 percentage points. The coefficient for `log_gdp_change` is -0.5296. This means that holding the change in the proportion of the elderly population in a country constant, when the change in $\log(\text{GDP per capita})$ increases by 1 unit (i.e., 172% cumulative GDP per capita growth from 2000 to 2019), then the change in mortality rates decreases by 0.5296 percentage points.

The p-value for `pop_65_change` is 0, so `pop_65_change` is still a reliable predictor. The p-value for `log_gdp_change` is 0.16. While this p-value decreased a lot from 0.298, the original value for the simple linear OLS we did earlier, it is still statistically insignificant.

The R-Squared is 0.384, which means that 38.4% of the variation in changes in death rates is explained by changes in the proportion of the population that is elderly and changes in $\log \text{GDP per capita}$. Including just `pop_65_change` in simple linear regression has an R-Squared of 0.377, and including just `log_gdp_change` in simple linear regression has an R-Squared of 0.006. So, including both predictors explains greater variation in the changes in mortality rates.

The adjusted R-Squared is 0.377, which means that 37.7% of the variation in changes in death rates is explained by changes in the proportion of the population that is elderly and changes in $\log \text{GDP per capita}$, after penalizing for added terms/model complexity. Including just `pop_65_change` in simple linear regression has an adjusted R-Squared of 0.374, and including just `log_gdp_change` in simple linear regression has an adjusted R-Squared of 0. So, the adjusted R-Squared did NOT meaningfully increase when adding the `log_gdp_change` term (only a 0.003 increase)!

Since the adjusted R-Squared did NOT meaningfully increase, adding the `log_gdp_change` predictor does not substantially increase the model's ability to explain variations in changes in mortality rates. Thus, we should discard `log_gdp_change` as a predictor in this multivariate model and revert to a simple linear regression with just `pop_65_change`.

Since there are 2 predictors in this multivariate regression, we could consider a ridge regression for improved results. The purpose of a ridge regression is to reduce multicollinearity between predictors to create a more stable and accurate model. But the correlation between `log_gdp_change` and `pop_65_change` is extremely low at 0.008515, so there is negligible multicollinearity between the variables. As such, there is minimal overlap in the ability of both of these predictors to explain the variation in changes in mortality rates, so a ridge regression is unnecessary and overcomplicates the model without enough benefit. Thus, we will not run a ridge regression.

Final Verdict

After running all these regressions, we find that the best model, relatively speaking, is **regression #3**, the simple linear regression with changes in the proportion of the elderly population (pop_65_change).

However, this model is not significant because the adjusted R-Squared is only 0.374. So, changes in the proportion of elderly populations only explains 37.4% of the variation for changes in mortality rates. Consequently, the model does NOT explain the majority of the remaining 62.6% of variation for the changes in mortality rates.

Omitted variable bias is the main reason the model fails to explain the majority of variation in mortality rate changes. This means that another predictor variable that strongly influences both the dependent variable (change in death rate) and one of the existing predictor variables (changes in proportion of elderly populations) is not in the model. Consequently, the predictor in the model (pop_65_change) already captures some of the explanatory power of the missing variable, thereby making it (pop_65_change) look less influential than it truly is.

The pop_65_change predictor is NOT just capturing the explanatory power of aging populations on changes in mortality rates. For instance, countries such as Germany have both a greater proportion of elderly populations AND better healthcare systems. A larger elderly population increases mortality rates, whereas better healthcare systems decrease mortality rates. So, the omitted variable is the quality of the healthcare system. Improvements in healthcare and the resulting decline in mortality rates might offset the increases in mortality rates coming from the aging population, thereby making it appear as though aging populations do not significantly contribute to changes in mortality rates. Consequently, we cannot conclude that changes in the proportion of elderly populations truly cause changes in mortality rates.

The aforementioned point is further exemplified by the difference in changes in mortality rates for low and high-growth countries in Graph 4. Low-growth countries experienced declines in mortality rates because the omitted variable, very high growth in healthcare system quality, outweighed the impact of the smaller growth in the elderly population. Conversely, very high-growth countries experienced increases in mortality rates because the stagnant healthcare system quality did not outweigh the impact of large growth in the elderly population.

Reflection

What worked well: Creating a correlation matrix to determine which predictors were most correlated with the change in mortality rates was effective because it enabled me to drop log_health_change from the model and avoid running unnecessary regressions. Testing multiple models (simple linear, polynomial, logarithmic, and multivariate linear) was useful because it confirmed that linear regression was the best fit for the pop_65_change, our most significant predictor.

What I would improve: The main way to improve the model is to incorporate different predictors to reduce confounding and omitted variable bias, such as healthcare system quality, because in reality, variables driving changes in mortality rates are multidimensional, so we cannot use just 1 variable to explain it.